UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	INTERACTIVE MULTIMEDIA
COURSE CODE	:	CSC413
EXAMINATION	:	NOV 2026 (MOCK)
TIME	:	3 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC413 mock paper (NOV 2026). Answers show the expected marking points; all questions are deterministic and paper-derived; no lecture slides are used.

2. Marking points show the expected answer and the allocation of marks; unresolved slots would remain visible.

3. Part A = 15 x 1 mark; Part B = 15 x 1 mark; Part C = 30 x 1 mark; Part D = 4 x 10 marks; total = 100.

4. Questions: 64. Slots: 72. Placeholders: 0. No LLM call. Questions and answers are for local study use only.

5. Structure derived from 15 unique CSC413 past papers; no past-paper text is copied.

FOR MARKERS' USE ONLY

This marking scheme consists of 6 printed pages

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PART A - ANSWER KEY

Q	Ans	Statement	Explanation
A1	TRUE	A storyboard is a visual plan of a multimedia sequence.	Storyboards show the planned flow.
A2	FALSE	Interlacing displays every line of a video frame at the same time.	Interlacing draws alternate lines.
A3	TRUE	Frame rate is the number of video frames displayed per second.	Frame rate is measured in fps.
A4	TRUE	Streaming allows media to begin playing before the whole file is downloaded.	Streaming delivers media continuously.
A5	TRUE	Path animation moves an object along a predefined route.	Path animation follows a motion path.
A6	FALSE	RAM is a type of permanent storage.	RAM is temporary working memory.
A7	TRUE	A scanner is an input device.	A scanner captures documents into the computer.
A8	TRUE	Sans serif fonts are generally preferred for screen display.	They are cleaner at small screen sizes.
A9	TRUE	Kerning adjusts the space between individual character pairs.	Kerning changes spacing for a specific pair.
A10	TRUE	A prototype is an early model of a product.	Prototypes test ideas before production.
A11	FALSE	A multimedia project always uses every media element.	A project selects the elements it needs.
A12	FALSE	A hyperlink can link only to another web page, not to a file.	Links can target files and other resources.
A13	TRUE	Authoring tools help developers organise and edit multimedia elements.	Authoring tools assemble media.
A14	FALSE	MIDI files store digitized audio samples.	MIDI stores musical instructions, not samples.
A15	TRUE	A higher sampling rate generally produces a more accurate recording.	More samples capture more detail.

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PART B - ANSWER KEY

Q	Ans	Term	Definition
B1	I	Multimedia	integration of text, graphics, audio, video and animation
B2	L	Virtual reality	simulated environment
B3	M	Keyframe	important frame in an animation
B4	J	Browser	software that displays web pages
B5	O	Linear multimedia	presentation without user control of sequence
B6	B	Frame rate	frames displayed per second
B7	G	Chroma key	replacing a colour background
B8	D	Tweening	generating in-between frames
B9	A	Sampling	converting sound into digital values
B10	C	Storyboard	visual sequence plan
B11	H	Hypermedia	linked multimedia elements
B12	E	Scanner	input device that digitises documents
B13	N	Plotter	output device for large drawings
B14	F	Card-based tool	authoring tool organised like pages or cards
B15	K	Cel animation	hand-drawn frame-by-frame animation

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PART C - ANSWER KEY

Q	Ans	Correct answer
C1	D	Another resource
C2	C	Permission to use a work
C3	C	PNG
C4	A	Pixels
C5	A	Drawing input
C6	B	Temporary working memory
C7	B	Display web pages
C8	D	Vector
C9	A	Smooth jagged edges
C10	C	A low-fidelity layout sketch
C11	B	Printed output
C12	A	Number of pixels in an image
C13	B	Modelling and rendering
C14	B	Analog sound to digital values
C15	C	Musical notes and instructions
C16	D	CMYK
C17	A	Non-linear
C18	A	Drawing each frame by hand
C19	B	Frequency
C20	A	Space between individual character pairs
C21	B	Multimedia with hyperlinks
C22	D	The user can control content and flow
C23	A	Merging of media and computing
C24	D	A cinema presentation
C25	C	An early model of a product
C26	D	Assemble multimedia elements
C27	D	Replace a colour background
C28	B	It combines visuals and sound
C29	B	Important frames in an animation
C30	C	Screen display

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PART D - WORKED ANSWERS

QUESTION 1

Topic: animation

You are required to animate a company logo. The logo depicts a planet orbiting the sun.

a) State TWO animation file formats suitable for this project.

GIF and MP4 are suitable; SWF is also acceptable.

(2 marks)

b) Describe the steps of cel animation and ONE advantage of using it.

Draw each frame, capture the frames in sequence, then play them back; it gives direct artistic control.

(4 marks)

c) Differentiate 2-D and 2.5-D animation with ONE example for each.

2-D uses flat drawings such as a cartoon; 2.5-D adds depth to flat artwork, such as parallax layers in a game.

(4 marks)

QUESTION 2

Topic: design project

A shopping complex requires a kiosk that gives customers location and retailer information.

a) State TWO important project team roles for completing the kiosk.

A project manager and an interface designer are important roles.

(3 marks)

b) Design a low-fidelity prototype for a pre-school courseware home page. The page must show topic menus, the course title and the school logo.

A simple sketch with a title, logo and large topic buttons is sufficient at low fidelity.

(3 marks)

c) Discuss TWO factors that lead to the wide use of mobile devices in games and shopping applications.

Portability and constant connectivity make mobile devices convenient for games and shopping.

(4 marks)

QUESTION 3

Topic: graphics

A product catalogue must show sharp images on screen and in print.

a) State TWO differences between bitmap and vector images.

Bitmap images use pixels and can pixelate; vector images use mathematical objects and scale cleanly.

(3 marks)

b) Describe TWO factors that affect image quality.

Resolution and colour depth both affect how much detail an image can show.

(3 marks)

c) Explain how colour model and resolution affect screen and print output.

Screens use RGB and print uses CMYK; print normally needs a higher resolution for sharp output.

(4 marks)

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QUESTION 4

Topic: authoring interactivity

A company wants an interactive product demonstration that runs on a kiosk and on the web.

a) State TWO features to consider when choosing authoring software.

Interactivity support and export formats are important features.

(2 marks)

b) Explain the interactivity level suitable for the demonstration.

A medium level with menus and buttons lets users explore without overwhelming them.

(3 marks)

c) Explain TWO authoring tool categories appropriate for this project.

Card-based tools suit structured screens; timeline-based tools suit animation and media sequencing.

(5 marks)

END OF MARKING SCHEME

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